07_duplicate_redundant_indexes

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

PURPOSE

- Detect indexes that duplicate or overlap existing ones.
- Every duplicate index adds overhead to INSERT, UPDATE and DELETE without adding read benefit.
- Identify and document which ones are safe to remove.

7.0 ENVIRONMENT CHECK

```
USE STN_Lab;
GO

-- ============================================================
==
-- 7.0 -- ENVIRONMENT CHECK
-- Estado actual de todos los indices en las tablas STN.
-- ============================================================
==

SELECT
    t.name                                              AS table_
name,
    i.index_id,
    i.name                                              AS index_
name,
    i.type_desc,
    i.is_unique,
```

```sql
    i.is_primary_key,
    STRING_AGG(
        CASE WHEN ic.is_included_column = 0 THEN c.name END,
', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal)      AS key_co
lumns,
    STRING_AGG(
        CASE WHEN ic.is_included_column = 1 THEN c.name END,
', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal)      AS includ
ed_columns
FROM sys.indexes i
JOIN sys.tables t
    ON i.object_id = t.object_id
JOIN sys.index_columns ic
    ON i.object_id = ic.object_id
   AND i.index_id  = ic.index_id
JOIN sys.columns c
    ON ic.object_id = c.object_id
   AND ic.column_id = c.column_id
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
GROUP BY t.name, i.index_id, i.name, i.type_desc, i.is_uniqu
e, i.is_primary_key
ORDER BY t.name, i.index_id;
GO
```

Results | Messages

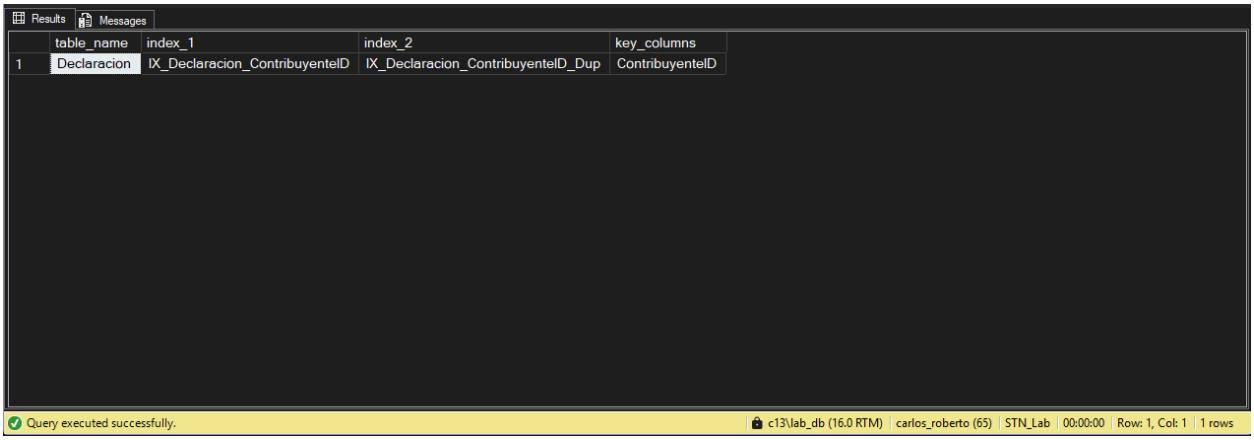
	table_name	index_id	index_name	type_desc	is_unique	is_primary_key	key_columns	included_columns
1	AuditoriaFiscal	2	PK_Auditoria	NONCLUSTERED	1	1	AuditorialD	NULL
2	Contribuyente	1	PK_Contribuyente	CLUSTERED	1	1	ContribuyenteID	NULL
3	Contribuyente	2	IX_Contribuyente_NIT	NONCLUSTERED	0	0	NIT	NULL
4	Declaracion	2	PK_Declaracion	NONCLUSTERED	1	1	DeclaracionID	NULL
5	Declaracion	5	IX_Declaracion_ContribuyenteID	NONCLUSTERED	0	0	ContribuyenteID	NULL
6	Declaracion	9	IX_Declaracion_Contribuyente_Periodo	NONCLUSTERED	0	0	ContribuyenteID, PeriodoFiscal	MontoDeclarado, Estado
7	Declaracion	10	IX_Declaracion_Periodo_Contribuyente	NONCLUSTERED	0	0	PeriodoFiscal, ContribuyenteID	MontoDeclarado, Estado
8	Declaracion	11	IX_Declaracion_Estado_Fecha	NONCLUSTERED	0	0	Estado, FechaPresentacion	ContribuyenteID, MontoDeclarado
9	Declaracion	12	IX_Declaracion_Fecha_Estado	NONCLUSTERED	0	0	FechaPresentacion, Estado	ContribuyenteID, MontoDeclarado
10	Pago	2	PK_Pago	NONCLUSTERED	1	1	PagoID	NULL

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (65) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 10 rows

7.1 DETECTAR INDICES DUPLICADOS EXACTOS

```
SELECT
    t.name                                                  AS table_
name,
    i1.name                                                 AS index_
1,
    i2.name                                                 AS index_
2,
    key_cols.key_columns
FROM sys.indexes i1
JOIN sys.indexes i2
    ON i1.object_id = i2.object_id
   AND i1.index_id  < i2.index_id
   AND i1.type_desc = i2.type_desc
JOIN sys.tables t
    ON i1.object_id = t.object_id
CROSS APPLY (
    SELECT STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal) AS key_columns
    FROM sys.index_columns ic
    JOIN sys.columns c
        ON ic.object_id = c.object_id
       AND ic.column_id = c.column_id
    WHERE ic.object_id = i1.object_id
```

```sql
      AND ic.index_id  = i1.index_id
      AND ic.is_included_column = 0
) key_cols
WHERE key_cols.key_columns = (
    SELECT STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal)
    FROM sys.index_columns ic
    JOIN sys.columns c
        ON ic.object_id = c.object_id
       AND ic.column_id = c.column_id
    WHERE ic.object_id = i2.object_id
      AND ic.index_id  = i2.index_id
      AND ic.is_included_column = 0
)
  AND t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
ORDER BY t.name;
GO
```



7.2 DETECTAR INDICES REDUNDANTES

```sql
SELECT
    t.name                                          AS table_
name,
    i_short.name                                    AS index_
```

```sql
redundante,
    i_long.name                                     AS index_
que_lo_cubre,
    short_cols.key_columns                          AS column
as_redundante,
    long_cols.key_columns                           AS column
as_cubridor
FROM sys.indexes i_short
JOIN sys.indexes i_long
    ON i_short.object_id = i_long.object_id
   AND i_short.index_id  <> i_long.index_id
   AND i_short.type_desc  = i_long.type_desc
JOIN sys.tables t
    ON i_short.object_id = t.object_id
CROSS APPLY (
    SELECT STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal) AS key_columns
    FROM sys.index_columns ic
    JOIN sys.columns c
        ON ic.object_id = c.object_id
       AND ic.column_id = c.column_id
    WHERE ic.object_id = i_short.object_id
      AND ic.index_id  = i_short.index_id
      AND ic.is_included_column = 0
) short_cols
CROSS APPLY (
    SELECT STRING_AGG(c.name, ', ')
        WITHIN GROUP (ORDER BY ic.key_ordinal) AS key_columns
    FROM sys.index_columns ic
    JOIN sys.columns c
        ON ic.object_id = c.object_id
       AND ic.column_id = c.column_id
    WHERE ic.object_id = i_long.object_id
      AND ic.index_id  = i_long.index_id
      AND ic.is_included_column = 0
) long_cols
```

```sql
WHERE long_cols.key_columns LIKE short_cols.key_columns + '%'
  AND i_short.is_primary_key = 0
  AND i_long.is_primary_key  = 0
  AND t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
ORDER BY t.name;
GO
```

Results | Messages

	table_name	index_redundante	index_que_lo_cubre	columnas_redundante	columnas_cubridor
1	Declaracion	IX_Declaracion_ContribuyenteID	IX_Declaracion_Contribuyente_Periodo	ContribuyenteID	ContribuyenteID, PeriodoFiscal
2	Declaracion	IX_Declaracion_ContribuyenteID	IX_Declaracion_ContribuyenteID_Dup	ContribuyenteID	ContribuyenteID
3	Declaracion	IX_Declaracion_ContribuyenteID_Dup	IX_Declaracion_ContribuyenteID	ContribuyenteID	ContribuyenteID
4	Declaracion	IX_Declaracion_ContribuyenteID_Dup	IX_Declaracion_Contribuyente_Periodo	ContribuyenteID	ContribuyenteID, PeriodoFiscal

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (65) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 4 rows

OBSERVACION 7.2

- Anotar índices redundantes encontrados.
- El index_redundante puede eliminarse si index_que_lo_cubre tiene las mismas o mas columnas líder.

7.3 INDICES SIN USO (DMV)

Índices que existen pero que el optimizador no ha usado desde el ultimo reinicio del servicio.

```sql
SELECT
    t.name                          AS table_name,
    i.name                          AS index_name,
    i.type_desc,
    ISNULL(us.user_seeks, 0)        AS user_seeks,
    ISNULL(us.user_scans, 0)        AS user_scans,
    ISNULL(us.user_lookups, 0)      AS user_lookups,
```

```sql
    ISNULL(us.user_updates, 0)      AS user_updates,
    s.used_page_count * 8           AS size_kb
FROM sys.indexes i
JOIN sys.tables t
    ON i.object_id = t.object_id
JOIN sys.dm_db_partition_stats s
    ON i.object_id = s.object_id
   AND i.index_id  = s.index_id
LEFT JOIN sys.dm_db_index_usage_stats us
    ON i.object_id = us.object_id
   AND i.index_id  = us.index_id
   AND us.database_id = DB_ID()
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
  AND i.index_id > 0
  AND i.is_primary_key = 0
ORDER BY ISNULL(us.user_seeks, 0) + ISNULL(us.user_scans, 0)
ASC;
GO
```

Results | Messages

	table_name	index_name	type_desc	user_seeks	user_scans	user_lookups	user_updates	size_kb
1	Contribuyente	IX_Contribuyente_NIT	NONCLUSTERED	0	0	0	0	4312
2	Declaracion	IX_Declaracion_ContribuyenteID	NONCLUSTERED	0	0	0	0	27856
3	Declaracion	IX_Declaracion_Estado_Fecha	NONCLUSTERED	0	0	0	0	55560
4	Declaracion	IX_Declaracion_Fecha_Estado	NONCLUSTERED	0	0	0	0	55672
5	Declaracion	IX_Declaracion_ContribuyenteID_Dup	NONCLUSTERED	0	0	0	0	27856
6	Declaracion	IX_Declaracion_Periodo_Contribuyente	NONCLUSTERED	1	0	0	0	55696
7	Declaracion	IX_Declaracion_Contribuyente_Periodo	NONCLUSTERED	2	0	0	0	55640

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (65) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 7 rows

OBSERVACION 7.3

- Indices con user_seeks = 0 y user_scans = 0 no han sido usados para lectura. Si user_updates es alto representan overhead puro en escritura sin beneficio en lectura.

- Candidatos a eliminacion previa validacion en produccion.

7.4 COSTO DE ESCRITURA POR INDICE

Cada indice adicional incrementa el costo de INSERT, UPDATE y DELETE. Esta query muestra el overhead real.

```
SELECT
    t.name                          AS table_name,
    i.name                          AS index_name,
    ISNULL(us.user_updates, 0)      AS write_operations,
    ISNULL(us.user_seeks, 0)
    + ISNULL(us.user_scans, 0)      AS read_operations,
    CASE
        WHEN ISNULL(us.user_seeks, 0)
           + ISNULL(us.user_scans, 0) = 0 THEN 'Solo escritur
a -- candidato a eliminar'
        WHEN ISNULL(us.user_updates, 0)
           > (ISNULL(us.user_seeks, 0) + ISNULL(us.user_scan
s, 0)) * 10
            THEN 'Escritura supera lectura 10x -- evaluar'
        ELSE 'Balance aceptable'
    END AS evaluacion
FROM sys.indexes i
JOIN sys.tables t
    ON i.object_id = t.object_id
LEFT JOIN sys.dm_db_index_usage_stats us
    ON i.object_id = us.object_id
   AND i.index_id  = us.index_id
   AND us.database_id = DB_ID()
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
  AND i.index_id > 0
  AND i.is_primary_key = 0
ORDER BY t.name, ISNULL(us.user_updates, 0) DESC;
GO
```

Results | Messages

	table_name	index_name	write_operations	read_operations	evaluacion
1	Contribuyente	IX_Contribuyente_NIT	0	0	Solo escritura -- candidato a eliminar
2	Declaracion	IX_Declaracion_ContribuyenteID	0	0	Solo escritura -- candidato a eliminar
3	Declaracion	IX_Declaracion_Contribuyente_Periodo	0	2	Balance aceptable
4	Declaracion	IX_Declaracion_Periodo_Contribuyente	0	1	Balance aceptable
5	Declaracion	IX_Declaracion_Estado_Fecha	0	0	Solo escritura -- candidato a eliminar
6	Declaracion	IX_Declaracion_Fecha_Estado	0	0	Solo escritura -- candidato a eliminar
7	Declaracion	IX_Declaracion_ContribuyenteID_Dup	0	0	Solo escritura -- candidato a eliminar

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (65) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 7 rows

7.5 CLEANUP

```
--Se creó para fines de ilustración de duplicación en el ambi
ente.
DROP INDEX IX_Declaracion_ContribuyenteID_Dup ON dbo.Declarac
ion;
```